Evolution, 2026, **0(0)**, 1–17
https://doi.org/10.1093/evolut/qpag127
Advance access publication 10 July 2026
Original Article

OXFORD

Modeling within-lineage evolution as a consequence of changes in the adaptive landscape across short and long timescales

Marion Thaureau and Kjetil Lysne Voje

Natural History Museum, University of Oslo, Oslo, Norway

Corresponding author: Natural History Museum, University of Oslo, Postboks 1172, Blindern 0318 Oslo, Norway. Email: marion.thaureau@nhm.uio.no

Abstract

The adaptive landscape has been suggested as a potential conceptual bridge between phenotypic evolution on generational and macroevolutionary timescales. However, this potential remains largely untapped due to a limited understanding of how it changes through time. Here, we assessed the dynamics of the adaptive landscape across different time intervals by analyzing phenotypic time series using multivariate models of adaptation. First, we examined whether a decrease in river water flow affected the optimal body mass of a salmon population over a few decades. Second, we explored whether temperature changes affected the optimal size of coccoliths in a species of coccolithophore across about a hundred thousand years in the Cretaceous. Finally, we analyzed the extent to which temperature changes affected the optimal shell size in a lineage of planktic foraminifera over a few million years during the Miocene. We find evidence that evolution induced by changes in the adaptive landscape, driven by environmental change, occurred in the salmon and foraminifer datasets, but we do not detect a causal link between the investigated paleo–proxies and size changes in the coccolithophore lineage. We argue that applying the same modeling framework to datasets covering different time spans not only offers a promising way to better connect evolutionary processes across timescales but also enables tests of adaptive hypotheses about within-lineage evolution.

Keywords: timeseries, Ornstein-Uhlenbeck process, adaptation-inertia framework, macroevolution, microevolution

Introduction

Significant progress has been made after the Modern synthesis in modeling generational evolutionary change in quantitative traits (Hansen & Houle, 2008; Lande, 1979; Lande & Arnold, 1983; Lynch & Walsh, 1998; Walsh & Lynch, 2018), but how to link this body of theory to explain phenotypic change on longer timescales remains a challenge (Arnold et al., 2001; Estes & Arnold, 2007; Gingerich, 2009; Hansen, 2012; Holstad et al., 2024; Rolland et al., 2023; Schluter, 2024; Uyeda et al., 2011). Although mutation, selection, migration, and genetic drift are universal mechanisms operating at all times in lineages, naïve extrapolation of microevolutionary models generally fails to predict observed evolution beyond a handful of generations (Estes & Arnold, 2007; Hansen, 2012). Because the trait dynamics captured in within-lineage time series are what eventually manifest as among-species differences, studying phenotypic evolution in lineages observed over a range of durations is one way to better connect our understanding of the evolutionary process across timescales. One challenge, however, is finding a modeling framework appropriate for data that spans orders of magnitude in time.

The adaptive landscape has been suggested to be a common ground for developing a better understanding of phenotypic evolution across different timescales (Arnold et al., 2001; Estes & Arnold, 2007; Hansen, 2012). The concept of the adaptive landscape was developed by Simpson (1944), where elevations in the landscape correspond to trait values correlated with higher fitness (i.e., more adapted traits). The curvature of the landscape where a population is currently situated determines the type and strength of selection acting on the population, as selection will always push populations towards peaks in the landscape. The adaptive landscape is a much-used model in both micro- and macroevolutionary studies, but the underlying assumption of the dynamics of the landscape often differs depending on the timescale investigated. While the landscape is commonly considered fixed on short timescales, it is often assumed to undergo drastic changes to explain macroevolutionary patterns. For example, Lande's (1976) impactful Ornstein-Uhlenbeck model of stabilizing selection and genetic drift represents phenotypic evolution on a fixed adaptive landscape, which is likely to adequately capture trait dynamics across a handful of generations in most cases. The model describes how a population ascends an adaptive peak and thereafter shows fluctuations around this peak, determined by both the curvature of the peak and the strength of genetic drift. Up to what timescales this model remains an adequate description of trait evolution is not clear. While patterns of fluctuating trait evolution around a steady state are common in the fossil record (Hunt, 2007; Hunt et al., 2015; Voje, 2016), the magnitude of these fluctuations is not predicted by a model of drift alone (Voje et al., 2018). A recent large-scale analysis of hundreds of phenotypic time series from the fossil record also found support for a model where lineages track a fluctuating (stationary) optimum (Holstad et al., 2024). Fi-

Received January 14, 2026; revisions received June 19, 2026; accepted June 29, 2026
Associate Editor: Laura Alencar; Handling Editor: Hélène Morlon

nally, and as pointed out by Hansen (1997, 2012), the almost immediate approach to the optimum predicted by Lande's selection-drift model (1976) is not compatible with the common finding that shared evolutionary history (i.e., phylogenetic signal) is a strong predictor of phenotypic evolution on macroevolutionary timescales (Ashton, 2004; Freckleton et al., 2002; Uyeda et al., 2011). In other words, the assumption of a fixed adaptive landscape over long timescales seems unlikely to reflect the reality of most biological systems, and might be violated even on rather short timescales.

In contrast to within-lineage analyses, the adaptive landscape is often allowed to change when applying Ornstein–Uhlenbeck (OU) models to among-species data in phylogenetic comparative studies (e.g., Hansen et al., 2008; Simon et al., 2025; Tsuboi et al., 2024; Voje & Hansen, 2013). When the OU process in these studies is interpreted as describing how the adaptive landscape itself changes over time, the approach is referred to as the adaptation-inertia framework (Bartoszek et al., 2012, 2023; Hansen, 1997, 2012; Hansen et al., 2008; Pienaar et al., 2026). The rationale for this interpretation is that the evolution of phenotypic differences between species likely involves changes in the adaptive landscape itself. Failure to perfectly and instantaneously track a shifting adaptive landscape manifests as inertia in trait evolution, which is often found when applying this framework to among-species data (e.g., Bartoszek et al., 2012; Hansen et al., 2008; Toljagić et al., 2018). To date, however, the adaptation-inertia framework has only been applied in a phylogenetic comparative context, and its extension to within-lineage phenotypic time series has not yet been explored. Consequently, relatively little is known about the lability of the adaptive landscape on timescales beyond a handful of generations (but see Hunt et al., 2008; Reitan et al., 2012; Voje et al., 2024). Because the framework allows optima either to change or to be fixed, applying it to within-lineage time series spanning a wide range of durations provides a single modeling approach that connects evolutionary dynamics across timescales through the process of adaptation.

Applying the adaptation-inertia framework to phenotypic time series can also aid in testing hypothesized drivers of observed change within evolving lineages. If, how and which environmental changes affect phenotypic evolution in single populations or species have been extensively studied and potential drivers of phenotypic evolution have been identified in numerous case studies; both in the fossil record (e.g., Hodell & Vayavananda, 1993; Hunt & Roy, 2006; Potts, 1998) and in contemporary taxa (e.g., Grant & Grant, 2002, 2006; Jensen et al., 2022; Losos et al., 1997; Meyer, 1987). Most approaches designed to test for drivers of change in the fossil record focus on correlated changes between variables of interest (e.g., Hunt et al., 2010). Such immediately correlated effects might be a good assumption for many biological systems where a change in one trait immediately leads to a change in a second trait (e.g., allometry), or where the response is geologically instantaneous given the resolution of the data. But there are also situations where changes in traits may be more delayed due to various factors, like genetic and functional evolutionary constraints. One of the strengths of the adaptation–inertia approach is its ability to disentangle causal signals from correlated changes among variables.

Here, we apply the adaptation-inertia framework by fitting multivariate OU models to three phenotypic timeseries of within-lineage evolution that vary by orders of magnitude in the time interval they cover. The first analysis considers whether the amount of river water flow affects the body mass across almost eight decades in a lineage of salmon, *Salmo salar* Linnaeus 1758 (data from Jensen et al., 2022) in the river Eira. River Eira in Norway used to host some of the largest salmon globally, but the population has, over the last century, seen a rapid decline in body mass. Jensen et al. (2022) argued that these notable changes in body mass are due to strong directional selection towards an optimum that changed due to the reduced water flow in the river as a consequence of the development of hydroelectric power plants. As a proof-of-concept, we reanalyze the data from Jensen et al. (2022) and investigate whether our OU time-series framework finds evidence for the claim that the average water volume measured from June to September during the spawning season affects the position of the adaptive peak for salmon body mass in the river.

The second analysis assesses drivers of body size change in the coccolithophore lineage *Biscutum constans* (Górka, 1957) Black, 1967 across approximately a hundred and sixty thousand years during the Cretaceous (phenotypic data from Bornemann & Mutterlose, 2006; isotopic data from Bornemann et al., 2005). One of the main debated drivers of the coccolith size variation is temperature, but many other environmental factors have been hypothesized to cause phenotypic change in the species *B. constans,* such as light, nutrient availability, oceanic acidification, or metal concentrations, as summarized by Bottini & Faucher (2020). We test whether, and how strongly, changes in temperature inferred from oxygen isotopes and variations in carbon isotopes affect the coccolith size optimum for *B. constans*.

The third analysis assesses whether changes in temperature (also estimated via oxygen isotopes) affect the adaptive landscape for size in the foraminifera *Globorotalia (Fohsella)* Bandy, 1972 lineage across 2.8 million years (Hodell & Vayavananda, 1993). Isotopes in planktic foraminifera have been shown to be reliable environmental proxies (e.g., Emiliani, 1955; Fairbanks et al., 1980, 1982; Ravelo & Hillaire-Marcel, 2007), and increase in δ^{18}O recorded in foraminifera shells from the middle Miocene is usually interpreted as the cooling of the oceanic surface layer due to the middle Miocene Climatic Transition (Flower & Kennett, 1994; Woodruff & Savin, 1989). However, while Hodell & Vayavananda (1993) measured oxygen isotopic ratios in several lineages of the same samples, they only found an increase in δ^{18}O in the calcitic shells of the genus *Globorotalia (Fohsella)*. As an explanation, they suggest a decrease in calcification temperature for this specific lineage rather than a fluctuation of continental ice volume. Based on the hypothesis that *Globorotalia (Fohsella)*'s size is sensitive to environmental changes, we test if and how strongly the temperature changes estimated by the oxygen isotopes and variation in carbon isotopes are influencing changes in the lineage's body length.

Applying multivariate OU models to phenotypic time series, where changes in one variable can predict changes in another (Granger causality; Granger, 1969) and where cause and effect can span a continuum from immediate to exceedingly slow, offers a principled way to connect phenotypic evolution across the timescale continuum. By applying the adaptation-inertia framework to three timeseries spanning vastly different timescales, our goal is twofold. First, we want to assess the potential of modeling the adaptive landscape as a way to understand within-lineage trait evolution in an attempt to unify our understanding of phyletic evolution across time. Second, we want to test specific hypotheses of adaptation in the analyzed lineages.

Materials and methods

The adaptation-inertia framework applied to single lineages

In this section, we briefly discuss the adaptation-inertia framework and how it can be interpreted when applied to evolutionary time series. For a more comprehensible introduction to the adaptation-inertia framework in phylogenetic comparative analyses, we refer the reader to Pienaar et al. (2026).

The goal of the adaptation-inertia approach is to understand how changes in the adaptive landscape affect trait evolution. This is done by modeling the evolution of the adaptive landscape itself. To be able to do this, the approach assumes that lineages are never far away from the adaptive peak they track the movements of. In a phylogenetic context, Hansen (1997) defined the average trait optimum to which a species within a niche or adaptive zone evolves towards as a "primary optimum." Species inhabiting the same adaptive zone likely have similar optima for a given trait, although not identical due to their unique biology. Conversely, differences in primary optima among species inhabiting different adaptive zones are considered positive evidence for systematic differences in their peaks' position in the adaptive landscape (sensu Simpson, 1953). The adaptation-inertia framework evaluates the extent to which species inhabiting different environments tend to evolve toward similar peaks, and hypotheses of adaptation can be assessed by investigating whether species living in different environments have a tendency to evolve towards different primary optima (Hansen, 1997). Rapid evolution of species towards distinctly different primary optima suggests that the environmental variable used to model the optima is important in explaining the evolution of the trait. However, traits are usually not only affected by selection towards a single peak. Factors like genetic drift, indirect selection of the trait due to genetic correlations with other traits under selection, and directional selection of the trait due to other unmeasured selective factors not explicitly modeled can all cause the trait to deviate from the primary optimum.

The simplest model that captures the above-mentioned properties is an Ornstein-Uhlenbeck process, which consists of the sum of a deterministic and a stochastic term. Within the adaptation-inertia framework, the deterministic term describes the evolution of the trait towards the primary optimum, while the stochastic term represents random changes in adaptive peaks unrelated to the primary optimum caused by a combination of unknown and unmeasured forces. Mathematically, the OU process for a single trait is described by the stochastic differential equation:

$$dy\,(t) = -\alpha\,(\mathrm{y} - \theta)\,dt + \sigma\,d\mathrm{W}, \tag{1}$$

where dy is the change in y over a time interval dt, θ is the primary optimum, α (≥ 0) determines the rate of adaptation toward the primary optimum, dW is short-hand for independent normally-distributed stochastic changes with mean zero and unit variance, over a unit of time (white noise), and σ is the instantaneous standard deviation of these changes (i.e., W is a Brownian-motion process). To quantify the average rate of change towards primary optimum, Hansen (1997) suggested that the α-parameter could be calculated as a half-life, $t_{1/2} = \ln(2)/\alpha$, which represents the expected time for the trait means to evolve half the distance from their ancestral state to a new primary optimum. A short half-life relative to the time span covered by the data means the expectation is rapid adaptation to the primary optimum, and vice versa for a long half-life. A half-life several times longer than the timespan covered by the data means the deterministic part of the OU process is negligible, and the trait dynamics become similar to a Brownian-motion process. The primary optimum can be parameterized by incorporating both categorical and continuous variables. The latter demands a multivariate implementation of the OU process (e.g., Bartoszek et al., 2012, 2023; Butler & King, 2009; Hansen et al., 2008; Mitov et al., 2019; Reitan et al., 2012), which allows for a more general multivariate comparative method. The multivariate version of the univariate OU process (eq. 1) is described by:

$$d\mathbf{y}\,(t) = -\mathbf{A}\,(\mathrm{y}\,(t) - \theta\,(\mathrm{t}))\,dt + \mathbf{\Sigma} d\mathbf{W}\,(\mathrm{t})\,, \tag{2}$$

where the numerical trait values ($\boldsymbol{y}$) and primary optimum ($\boldsymbol{\theta}$) are vectors, and $\mathbf{W}$ corresponds to a multidimensional Brownian motion. The stochastic movement $\mathbf{\Sigma}$, and the rate of adaptation $\mathbf{A}$ take the form of matrices. Different parameterizations of the $\mathbf{A}$ matrix make it possible to assess a range of evolutionary hypotheses, including whether changes in one variable are causing changes in another. The latter situation is interpreted within the adaptation-inertia framework as the predictor variable affecting the position of the optimum toward which the response trait evolves. An $\mathbf{A}$ matrix with only diagonal elements represents an OU process where each variable changes independently toward its own primary optimum, whereas a non-zero off-diagonal element indicates that changes in one variable affect the primary optimum of another. For example, changes in two variables y_1 and y_2 (e.g., y_1 = phenotypic trait and y_2 = environmental variable) are described by a multivariate OU process in two dimensions (see also Figure 1):

$$d\begin{bmatrix} y_1\,(t) \\ y_2\,(t) \end{bmatrix} = -\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\left(\begin{bmatrix} y_1\,(t) \\ y_2\,(t) \end{bmatrix} - \begin{bmatrix} \theta_1\,(t) \\ \theta_2\,(t) \end{bmatrix}\right)\mathrm{d}t + \mathbf{\Sigma} d\mathbf{W}\,(\mathrm{t})\,, \tag{3}$$

where non-zero a_{12} means changes in y_2 affect the primary optimum of y_1 (Granger causality), and vice versa for a_{21}. Changes in y_1 are therefore both a function of the distance of y_1 to θ_1, and the distance of y_2 to θ_2, since y_1 is affected by changes in y_2 (stochastic variation is added to both variables):

$$dy_1\,(t) = -a_{11}\,(y_1\,(t) - \theta_1\,(t)) - a_{12}\,(y_2\,(t) - \theta_2\,(t)) + \mathbf{\Sigma} d\mathbf{W}\,(\mathrm{t})\,, \tag{4}$$

How changes in y_2 affect the primary optimum of y_1 can be interpreted as a linear regression obtained after reordering expression (4):

$$dy_1\,(t) = -a_{11}\left[(y_1\,(t) - \theta_1\,(t)) - \frac{a_{12}}{a_{11}}\,(y_2\,(t) - \theta_2\,(t))\right] + \mathbf{\Sigma} d\mathbf{W}\,(\mathrm{t})\,, \tag{5}$$

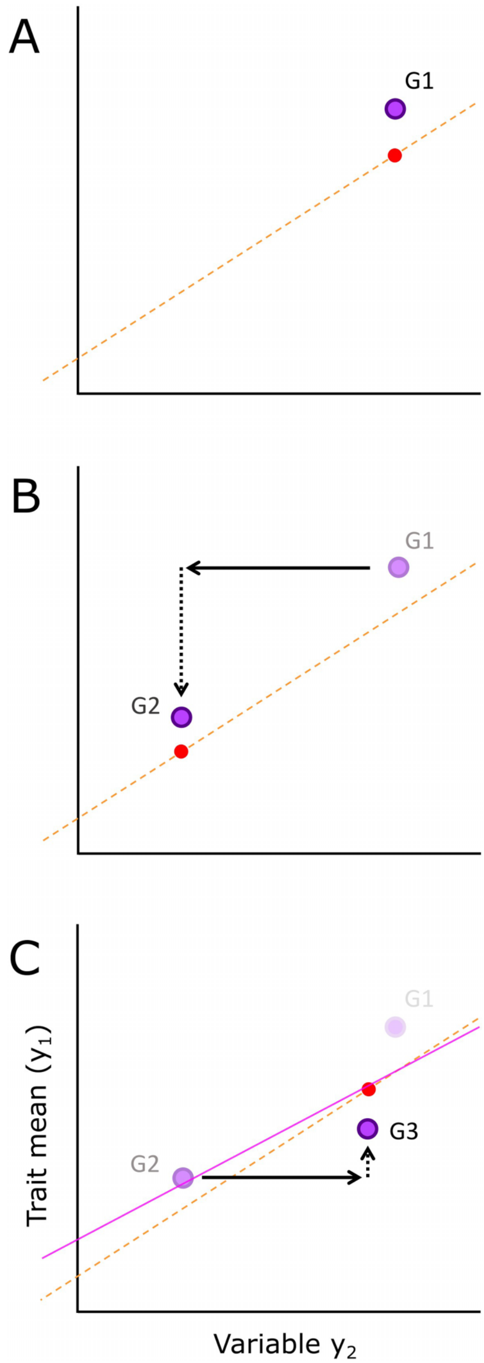

Figure 1. Optimal and observed regressions. Panels A, B, and C show how the mean of a trait y_1 and a variable y_2 (e.g., phenotypic or environmental variable) change across 3 generations within a lineage. The outlined dots indicate the empirical mean of the y_1 and y_2 at each point in time, here for generations G1, G2, and G3. The dashed line is the optimal regression, and the dot along the dashed line is the optimal trait value for a given value of y_2 in each generation (the optimal trait value is moving along the optimal regression as a function of changes in y_2). The closer y_1 is to the optimum, the better adapted the lineage is at that time. Horizontal arrows (plain) indicate a change in y_2-value, causing the optimal trait value to change (i.e., the position of the adaptive peak has changed). Vertical arrows (dashed) reflect changes in the trait while it evolves (adapts) towards the optimum (dot along the dashed line). A lag in the adaptive process exists if the trait is not able to evolve immediately to the optimal trait value. This lag causes a difference between the optimal regression and the observed regression of the trait y_1 and the variable y_2. The plain line in panel C is the evolutionary or observed regression, based on the three empirical data points.

where the optimal value of trait y_1 changes with a slope of $\frac{a_{12}}{a_{11}}$ due to changes in y_2. This slope represents the optimal regression (Figure 1, see also Pienaar et al., 2026). If a_{12} is null, the slope of the optimal regression between y_1 and y_2 is 0, and the primary optimum of y_1 is θ_1. The optimal regression line represents the optimal relationship between the two variables and would be the relationship between y_1 and y_2 if the adaptation of y_1 towards the optimal state is immediate with no lag, i.e., it represents the relationship between the optimum and the trait free of constraints (see e.g., Hansen et al., 2008). Any lag or inertia in the ability of the trait to evolve towards the optimal state makes the empirical relationship between the trait and the variable, the observed or evolutionary regression, less steep compared to the optimal relationship (Figure 1C). Quantifying the ability of the trait to evolve towards the primary optimum, therefore, allows us to characterize the lag in adaptation towards the primary optimum.

The stochastic influence on trait evolution is given by the $\boldsymbol{\Sigma}$ matrix. For a diagonal $\boldsymbol{\Sigma}$ matrix, the changes in each variable are subject to independent stochastic influences. For instance, the size of stochastic changes in y_1 is noted as σ_{11}, and the amplitude of variation when the trait has reached the optimum can be quantified using the formula of equilibrium variance $\sigma_{11}/(2*\alpha_{11})$ (Hansen et al., 2008). Any non-zero off-diagonal elements in the $\boldsymbol{\Sigma}$ matrix represent the covariance of the stochastic changes in the variables. Potential sources of such correlated random changes in two phenotypic variables can be shared developmental constraints or correlated evolution due to pleiotropy or linkage disequilibrium with other unmeasured traits under selection. When recovered between a phenotypic and an environmental variable, correlated random changes could indicate that a third variable is causing both to change in a correlated fashion.

We argue that the interpretation of the multivariate OU models within the adaptation-inertia framework is largely the same whether applied to time series or to phylogenetic comparative data, with a few exceptions. First, we use *primary optimum* in the same deterministic sense as in phylogenetic comparative analyses: it is the fixed value θ under independent adaptation, or the optimal regression $\theta(x)$ under dependent adaptation, and is distinct from the secondary, stochastic peak movements captured by the Σ term. What differs is its reference: here, the primary optimum is the systematic optimum the lineage tracks through time, rather than the average optimum across species in an adaptive zone. Under the assumption that the primary optimum does not change, variation in trait mean after the lineage has reached the estimated primary optimum is then due only to unmeasured selective factors (e.g., selection towards secondary optima) and drift captured by the stochastic component of the OU process, together with measurement and estimation error in all variables. Second, fitting a model with only a single fixed primary optimum for all species in a phylogenetic comparative analysis is a way to estimate the amount of phylogenetic effect in the data (Hansen et al., 2008), but it is equivalent to estimating the evolution towards a fixed primary optimum when analyzing time-series data.

Time series analyzed in this study

We analyzed phenotypic evolution in three different taxa spanning three different timescales using the adaptation-inertia framework. The first dataset consists of the body mass of a lineage of *Salmo salar* (in kg) recorded across almost eight decades in the river Eira (number of observations across time = 69; average sample size per data point = 117; data from Jensen et al., 2022), along with the water flow volume recorded between June and September. The second dataset contains the coccolith length of the Cretaceous coccolithophore lineage *B. constans* measured in micrometers across 160,000 years (number of observations across time = 50; sample size per datapoint = 60; data from Bornemann & Mutterlose, 2006). Bulk oxygen and carbon isotopes have also been measured in the samples (Bornemann et al., 2005). The third dataset includes changes in test length in the planktic Miocene foraminifera *Globorotalia (Fohsella)*. The lineage is considered a chronospecies, with six different species being taxonomically defined due to their gradual morphological changes over time (Fleisher, 1974). They occur successively in the stratigraphic section with some overlap during the hypothesized speciation events. Following Hodell & Vayavananda (1993), we averaged the foraminifera length and isotopic values across species when more than one species of the same genus was found in the same time interval. Body length is measured in micrometers (number of observations across time = 114; sample size per datapoint = 60; data from Hodell & Vayavananda, 1993). The dataset also contains carbon and oxygen isotopes measured on the lineage's shells, which we use as paleoenvironmental proxies.

The trait data in each of the three datasets comprises mean, variance, sample size, and absolute time for each observation. We analyzed changes in trait means on a log-scale, as we aim to investigate proportional changes in size across time in all three time series. We also log-transformed water flow to assess the effect of proportional changes in water volume in the river on the position of the size optimum in salmon. We did not log-transform the isotopic data as these are not on a ratio scale (Houle et al., 2011; Voje et al., 2023). We used the following R packages when analyzing the data: dplyr v.1.1.4 (Wickham et al., 2023), evoTS v.1.0.3 (Voje, 2023), and paleoTS v.0.6.2 (Hunt, 2006); and when visualizing the results: ggbreak v.0.1.6 (Xu et al., 2021), ggplot2 v.4.0.1 (Wickham, 2016), gridExtra v.2.3 (Auguie & Antonov, 2017), and scales v.1.4.0 (Wickham et al., 2025). Data, complete scripts, and results are available in the GitHub repository.

Fitting multivariate OU models

Multivariate OU models were fitted to each of the three datasets using the package evoTS v. 1.0.3 in R (Voje, 2023). All fitted models operate under the assumption that the sample means in an ancestral-descendant sequence possess a joint distribution that is multivariate normal. This distribution is characterized by an expected mean vector and covariance matrix, which are functions of each model's parameters, the time intervals between samples in the sequence, and the sampling variances of the trait means for each sample. Given the multivariate normality assumption of the sample means, their expected distribution is determined by their first, second, and mixed second-order (covariance). The expected distribution of differences between samples in a time series facilitates the computation of likelihoods for a set of parameter estimates when fitting a given model of evolution to an empirical dataset. The time from the first to the last sample in the time series was normalized to unit length prior to fitting the models, to enhance parameter estimation and facilitate the interpretation of model parameters. The estimation (sampling) error of sample means contributes to the variance among sample means. This estimation error was accounted for by adding the population sample variance divided by the number of specimens measured in that sample to the diagonal of the variance-covariance matrix (Hunt, 2006). We also account for an uncertainty in the mean of environmental variables, equal to 5% for the water flow, and 0.1% for both oxygen and carbon isotopic ratios. Note that using an OU process to describe an environmental time series does not imply biological adaptation or selection acting on the environment. It simply provides a statistical description of temporal variation, allowing us to assess whether the variable exhibits mean-reverting behavior around a long-term value or behaves more like an unbiased random walk. OU processes are used to model environmental data in various fields, such as in climatology or hydrology (e.g., Brody et al., 2002; Reitan & Petersen-Øverleir, 2011).

The stochastic changes $\mathbf{\Sigma}$ and the rate of adaptation $\mathbf{A}$ take the form of matrices in the multivariate OU model. In evoTS v.1.0.3, $\mathbf{\Sigma}$ is a positive semi-definite matrix, and $\mathbf{A}$ is a matrix whose eigenvalues are not required to have positive real parts. We parameterized the $\mathbf{A}$ and $\mathbf{\Sigma}$ matrices in different ways to assess the relative support for different hypotheses in each of the three datasets. Those hypotheses can be classified into three categories of models based on how the $\mathbf{A}$ matrix is parameterized (Figure 2, see also Bartoszek et al., 2023, and Voje, 2023 for similar classifications). Random evolution (Figure 2A) designates models where all the variables change according to an unbiased random walk (URW, only zero entries in the $\mathbf{A}$ matrix). Independent adaptation (Figure 2B) means a trait evolves toward a fixed optimum (the $\mathbf{A}$ matrix has only non-zero elements on the diagonal). Dependent adaptation (Figure 2C) describes models of Granger causality, where a trait evolves toward a changing optimum, and where variation in that optimum is caused by changes in at least one of the other investigated variables (i.e., at least one off-diagonal element in the $\mathbf{A}$ matrix is non-zero). In these three categories, stochastic variation among variables can be independent or correlated (diagonal or full $\mathbf{\Sigma}$ matrix). In models of independent and dependent adaptation, we allowed the environmental variable to either follow an unbiased random walk or an OU process towards a fixed optimum.

Time series analyses can potentially confuse a shared temporal trend with evidence of a causal relationship between variables. We address this concern in two ways. First, our set of fitted models always includes models where variables follow unbiased random walk processes. Simulations show that when both causal OU and unbiased random walk models are fitted to data generated from two independent trending variables, the causal model is rarely selected as the best-fitting model (1 out of 100 simulations; see Table S1 in Supplementary material). Second, we conduct a model recovery analysis for any empirical dataset where we find positive evidence for dependent adaptation. Data are simulated from the maximum likelihood parameter estimates and refitted to verify that our modeling framework is able to recover the correct model structure.

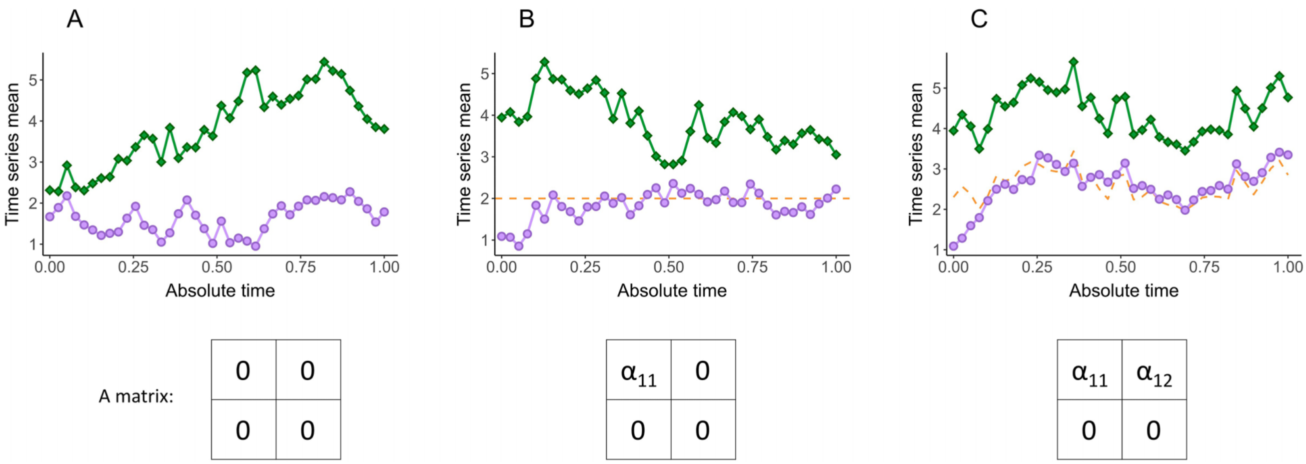

Figure 2. Three categories of multivariate models of phenotypic evolution. The panels display changes in the means of two time series across time (circle = phenotypic trait y1, diamond = environmental variable y2), exemplifying random change, independent adaptation, and dependent adaptation. The primary optimum is represented by an orange dashed line (B and C). The matrices represent examples of how the A matrix can be parametrized for each category of models. In all three examples, the environmental variable is following an unbiased random walk. (A) Random evolution: the trait is evolving randomly (unbiased random walk process); it is not tracking an optimum. (B) Independent adaptation: the trait is evolving independently of the environmental variable and evolves towards a fixed optimum. (C) Dependent adaptation: the trait is evolving toward a moving optimum that is driven by changes in the environmental time series.

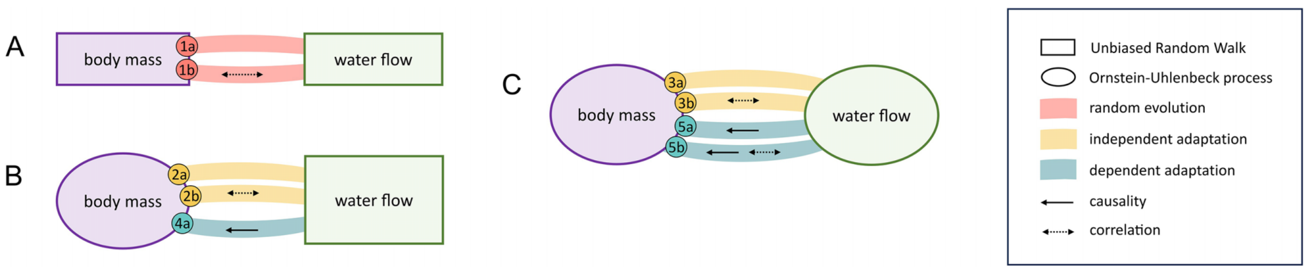

Figure 3. Models fitted to the salmon dataset. (A) Salmon body mass and water flow follow unbiased random walk (URW) processes. (B) Salmon body mass is evolving as an Ornstein-Uhlenbeck (OU) process, while water flow follows an URW process. (C) Both salmon body mass and water flow follow OU processes. Except for model 4a, each model is fitted in two variants: one with a correlation between the stochastic changes in the two variables and one without.

Salmon body mass

We fitted a total of nine different models to the salmon dataset. One set of models (Figure 3A) allows both variables to follow an unbiased random walk (i.e., no deterministic part in the OU process). We ran versions without (model 1a) and with correlated random changes (model 1b) between the variables (i.e., non-zero off-diagonal elements in the $\mathbf{\Sigma}$ matrix). A second set of models (Figure 3B) allows the salmon body size to evolve as an OU process, while the water flow changes according to a random walk. We ran both versions with absence (model 2a) and presence of correlated random changes (model 2b), and a version where the water flow did affect the optimum of the salmon body size (dependent adaptation-model 4a). Finally, a third set of models (Figure 3C) allows both salmon body size and water flow to vary as OU processes. Independent adaptation versions of this last set were run with the absence (model 3a) and presence of correlated changes (model 3b), as well as dependent adaptation versions where water flow affects the optimum of the salmon body mass without (model 5a) and with correlated random changes (model 5b).

Coccolithophore and foraminifera body length

We fitted the same twelve models to the datasets on changes in coccolith and foraminifera test size. In the first set (Figure 4A), all three variables change according to an unbiased random walk without (model 1a) and with correlated random stochastic changes (model 1b). In the second set of models (Figure 4B), the size trait evolves as an OU process towards a fixed optimum (independent adaptation), while the environmental proxies change according to an unbiased random walk. In this set, we investigate both the absence (model 2a) and the presence of correlated random changes among all three variables (model 2b). The trait and the two environmental variables follow OU processes in the third set of models (Figure 4C). Two of the models in the third set describe independent adaptation in the trait without (model 3a) and with correlated random changes among the three variables (model 3b). Six of the models in the third set describe dependent adaptation without and with correlated random changes among all variables, where the oxygen isotopes (models 4a and 4b), carbon isotopes (models 5a and 5b), or both variables influence the trait optimum (models 6a and 6b).

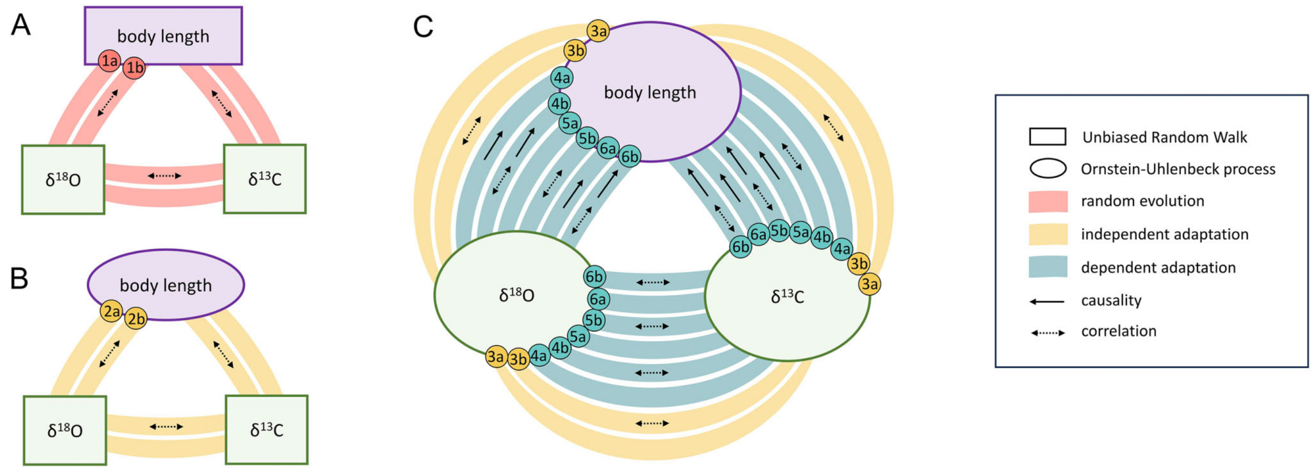

Figure 4. Models fitted to the coccolith and foraminifera datasets. (A) Both the phenotypic trait and the environmental proxies follow unbiased random walk (URW) processes. (B) The phenotypic trait is evolving as an Ornstein Uhlenbeck (OU) process while the environmental proxies follow an URW process. (C) Both the phenotypic trait and the environmental proxies follow an OU process. Each model is fitted in two variants: one with a correlation between the stochastic changes in the two variables and one without.

Fitting multivariate OU models involves searching a potentially multi-peaked likelihood space (e.g., Bartoszek et al., 2023; Voje, 2023). For each model we fitted, we let the search algorithm start from 100 different starting points in parameter space to extensively cover the log-likelihood landscape. Model fit was evaluated using the sample size adjusted Akaike Information Criteria (AICc) (Akaike, 1974). Alternative models are considered a good fit when their AICc falls into the 5-unit range close to the best model (Burnham et al., 2011).

Results

Salmon body mass

The best model for the salmon body size data is a model where both salmon body size and river water flow change according to OU processes, with changes in water flow affecting the primary optimum of salmon size (model 5a, Table 1). The estimated parameters in the **A** matrix give a slope of α_{12}/α_{11} = 0.98 for how changes in water flow affect the primary optimum of salmon size, i.e., a 10% change in water flow predicts a 9.8% change in the primary optimum for salmon size (Figure 5). The evolution of the salmon body mass towards the primary optimum is very rapid (half-life = $\ln(2)/\alpha_{11}$ = 1.2% of the time covered by the time-series = 0.91 years) with a decrease of the average body mass of the population from 11.06 kg to 3.06 kg over 76 years. The magnitude of the random changes in the trait (σ_{11}) is 7.89 log (gram^2), which gives an equilibrium variance of $\sigma_{11}/(2*\alpha_{11})$ = 0.07 log(gram^2) when the trait is in the proximity of the primary optimum. In our model recovery analysis, in which data were simulated from the maximum likelihood estimates of the best-fitting model and refitted, we found that the correct dependent adaptation model was recovered in 87 of 100 simulations, indicating that the empirical signal is strong enough for our modeling framework to reliably identify dependent adaptation in the salmon data (Supplementary material, Table S2).

Only model 5b has a fit comparable to that of the best model 5a (Table 1). It describes a very similar relationship between the salmon body mass and the water flow as the best model (Figure S1): the decrease in water flow strongly affects variation in the primary optimum of the salmon body mass (optimal slope = 1.12; half-life = 0.80 year), and the stochastic variation also behave in a very similar fashion compared to the best model (size of random changes = 7.71 log(gram^2); equilibrium variance = 0.06 log(gram^2). Model 5b differs from 5a as it allows for a correlation in the random changes in the two variables, which is estimated to be negative (correlation = -0.15).

Coccolith length

The model that best describes changes in the coccolith data (Table 2) is a model where all three variables change towards separate fixed states, with correlated stochastic changes (model 3b). According to this model, the coccolith length is not influenced by changes in either of the two environmental variables and evolves with a half-life of 159 ky (Figure 6). The size of the random changes in the trait is $1.58*10^{-5}$ log(µm^2), translating into an equilibrium variance of $1.19*10^{-5}$ log(µm^2) in the trait mean, implying a small effect of the stochastic changes on the trait dynamics. These stochastic changes are highly and negatively correlated with the stochastic variation in the oxygen and carbon isotopes (trait-oxygen = -0.78, trait-carbon = -0.96).

The only model showing a similar fit relative to the best model (model 4b, ΔAICc = 4.94, Figure S2) indicates that changes in the oxygen isotopic ratio leads to changes in the optimal size of the coccolithophore lineage, but the optimal regression slope is 0.01, meaning a 10% change in the oxygen isotopes would lead to 0.1% changes in the coccolith body size, which can be considered a very weak effect size. The stochastic changes show a very similar dynamic compared to the best model ($7.45*10^{-3}$ log(µm^2); equilibrium variance = $3.34*10^{-3}$ log(µm^2)). Model 4b also supports a strong negative correlation between the stochastic changes in the time series (trait-oxygen = -0.97, trait-carbon = -0.71). The weak effect of changes in the oxygen isotopic ratio on the optimal size in the coccolithophore lineage means the second-best model is very similar in interpretation to the best-fitting model.

Table 1. Model fit and parameter estimates for the salmon dataset.

Model type	Model *(k)*	ΔAICc/logL	Ancestral value ($z_{0.1}, z_{0.2}$)	Optimum (θ_1, θ_2)	A matrix: Diagonal (α_{11}, α_{22})	A matrix: Off-diagonal α_{12}	Σ matrix: Diagonal (σ_{11}, σ_{22})	Σ matrix: Off-diagonal σ_{12}
Random evolution	Model 1a *(4)*	23.19/-45.91	(9.09, 4.48)	(NA, NA)	(0, 0)	0	(7.01, 9.68)	0
	Model 1b *(5)*	25.3/-45.8	(9.09, 4.48)	(NA, NA)	(0, 0)	0	(6.98, 9.67)	0.51
Independent adaptation	Model 2a *(6)*	20.51/-42.2	(9.09, 4.48)	(8.39, NA)	(15.03, 0)	0	(7.92, 9.68)	0
	Model 2b *(7)*	22.81/-42.11	(9.09, 4.48)	(8.39, NA)	(15.01, 0)	0	(7.89, 9.67)	0.51
	Model 3a *(8)*	13.07/-35.96	(9.1, 4.47)	(8.33, 3.67)	(13.97, 30.61)	0	(7.89, 12.06)	0
	Model 3b *(9)*	13.86/-35.03	(9.09, 4.48)	(8.39, 3.68)	(18.43, 32.77)	0	(7.94, 12.69)	1.54
Dependent adaptation	Model 4a *(7)*	17.53/-39.47	(9.09, 4.48)	(-3.43, NA)	(32.03, 0)	-13.05	(8.65, 10.02)	0
	Model 5a *(9)*	0/-28.1	(9.06, 4.48)	(8.33, 3.65)	(57.33, 24.02)	-56.24	(7.89, 11.09)	0
	Model 5b *(10)*	0.42/-26.94	(9.1, 4.48)	(8.39, 3.72)	(65.93, 31.07)	-73.56	(7.71, 12.23)	-1.44

Notes: k is the number of parameters in the model. ΔAICc is the difference in AICc score relative to the best model. z0.1 and z0.2 correspond to the ancestral values of the trait and the water flow, and $\theta 1$ and $\theta 2$ are the optimum parameters in the OU process for each of the two variables. $\alpha 11$ and $\alpha 22$ represent the diagonal elements of the A matrix, while the off-diagonal element is $\alpha 12$. The diagonal elements of the Σ matrix ($\sigma 11$ and $\sigma 22$) are the stochastic variance, while $\sigma 12$ is the covariance of those stochastic changes. Standard errors of parameter estimates are available in the Supplementary material (Table S3).

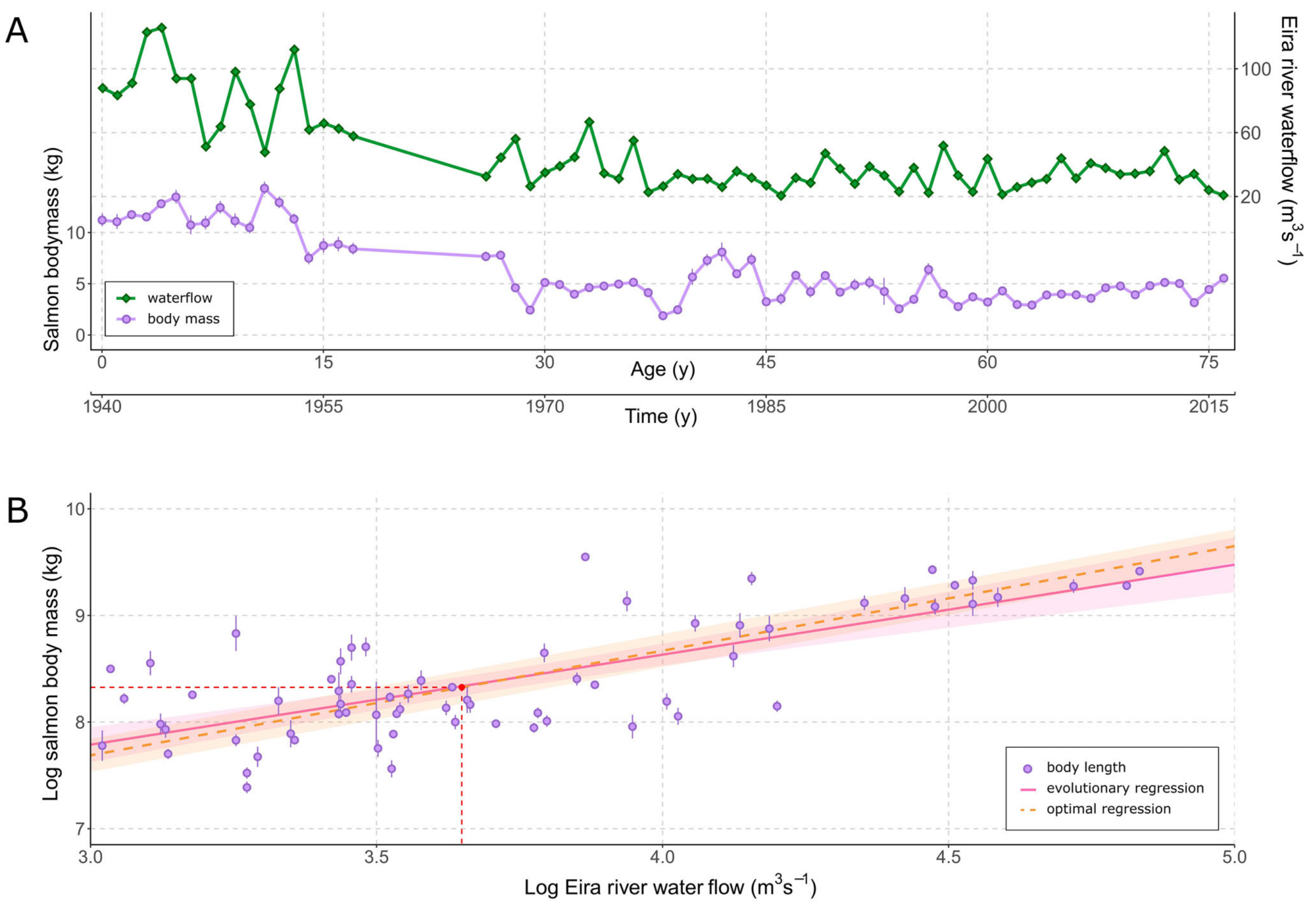

Figure 5. Time series and regressions of the best model (model 5a) for the salmon dataset. (A) Changes in mean salmon body mass (circles) and water flow (diamonds) recorded each year between June and September in the river Eira (data from Jensen et al., 2022). (B) Optimal (—) and evolutionary (OLS) regression (—) of log salmon body mass on log river water flow. The slope of the optimal regression is 0.98, while the evolutionary regression is 0.84. The intercept for the optimal regression is calculated as mean(log body size)–0.98∗mean(log water flow). The variance explained by the optimal regression is R2 = 53%, while it is 54% for the evolutionary regression. The plain dot on the optimal regression line represents the values the two time series converge towards if the stochastic parts of the two OU processes were nonexistent in both variables.

A consequence of an OU process with a substantial half-life can be inaccurate estimates of optima (e.g., Grabowski et al., 2023). Most of the estimated model parameters for the coccolithophore dataset also have high standard errors (Figure 7B, C; Table S4), likely due to the challenge in precisely defining the weak effect of the environmental variables on the trait optimum. In addition, despite extensive exploration of starting values, one of the models (model 6a) we

Table 2. Model fit and parameter estimates for the coccolithophore dataset.

Model type	Model *(k)*	ΔAICc/logL	Ancestral value ($z_{0.1}, z_{0.2}, z_{0.3}$)	Optimum ($\theta_1, \theta_2, \theta_3$)	A matrix: Diagonal ($\alpha_{11}, \alpha_{22}, \alpha_{33}$)	A matrix: Off-diagonal (α_{12}, α_{13})	Σ matrix: Diagonal ($\sigma_{11}, \sigma_{22}, \sigma_{33}$)	Σ matrix: Off-diagonal ($\sigma_{12}, \sigma_{13}, \sigma_{23}$)
Random evolution	Model 1a *(6)*	16.38/75.8	(1.27, -4.01, 1.59)	(NA, NA, NA)	(0, 0, 0)	(0, 0)	(0, 4.56, 3.47)	(0, 0, 0)
	Model 1b *(9)*	6.21/85.16	(1.27, -4, 1.59)	(NA, NA, NA)	(0, 0, 0)	(0, 0)	(0.01, 4.72, 3.55)	(-0.06, -0.06, 2.45)
Independent adaptation	Model 2a *(8)*	14.3/79.62	(1.28, -4.01, 1.59)	(1.23, NA, NA)	(2463.16, 0, 0)	(0, 0)	(5.47, 4.56, 3.47)	(0, 0, 0)
	Model 2b *(11)*	5.78/88.6	(1.27, -4, 1.59)	(1.14, NA, NA)	(0.89, 0, 0)	(0, 0)	(0, 4.5, 3.43)	(-0.02, -0.01, 2.25)
	Model 3a *(12)*	5.01/90.72	(1.27, -4.01, 1.59)	(1.13, -4.05, 1.6)	(0.78, 35.4, 25.25)	(0, 0)	(0, 6.82, 4.68)	(0, 0, 0)
	Model 3b *(15)*	0/99.07	(1.27, -4.01, 1.58)	(1.12, -4.06, 1.56)	(0.66, 36.48, 22.23)	(0, 0)	(0, 6.46, 4.05)	(-0.01, -0.01, 2.88)
Dependent adaptation	Model 4a *(13)*	8.6/90.77	(1.27, -4.01, 1.59)	(1.12, -4.05, 1.59)	(0.71, 33.05, 21.69)	(0, 0)	(0, 6.57, 4.36)	(0, 0, 0)
	Model 4b *(16)*	4.94/98.79	(1.28, -4, 1.6)	(1.16, -4.04, 1.62)	(1.12, 30.71, 29.22)	(-0.02, 0)	(0, 5.98, 5.3)	(-0.01, -0.01, 3.03)
	Model 5a *(13)*	8.66/90.74	(1.27, -4.01, 1.59)	(1.11, -4.05, 1.59)	(0.65, 33.86, 25.38)	(0, 0)	(0, 6.68, 4.66)	(0, 0, 0)
	Model 5b *(16)*	5.22/98.64	(1.29, -4, 1.58)	(1.22, -4.05, 1.58)	(10.65, 31.25, 20.55)	(0, -0.06)	(0.02, 6.45, 4.29)	(-0.12, -0.07, 3.06)
	Model 6a† *(14)*	12.68/90.67	(1.28, -4.01, 1.59)	(1.14, -4.05, 1.61)	(0.95, 36.36, 23.25)	(-0.03, -0.02)	(0, 6.92, 4.46)	(0, 0, 0)
	Model 6b *(17)*	8.99/99.08	(1.29, -4.01, 1.58)	(1.22, -4.03, 1.59)	(9.52, 29.55, 26.93)	(1.79, -1.87)	(0, 6.19, 4.63)	(-0.07, -0.1, 2.98)

Notes: k is the number of parameters in the model. ΔAICc is the difference in AICc score relative to the best model. z0.1, z0.2, and z0.3 correspond to the ancestral values of the log size trait, oxygen, and carbon isotope, and $\theta 1$ (coccolith length), $\theta 2$ (oxygen isotope), and $\theta 3$ (carbon isotope) are the optimum parameters in the OU process for each of the three variables. $\alpha 11$, $\alpha 22$, and $\alpha 33$ represent the diagonal elements of the A matrix, while the off-diagonal elements are $\alpha 12$ and $\alpha 13$. In the Σ matrix, $\sigma 11$, $\sigma 22$, and $\sigma 33$ are the stochastic variances in the size trait, the oxygen isotope, and the carbon isotope, respectively, while $\sigma 12$, $\sigma 13$, and $\sigma 23$ represent the covariances of the random changes in these variables. Standard errors of parameter estimates are available in the Supplementary material (Table S4).

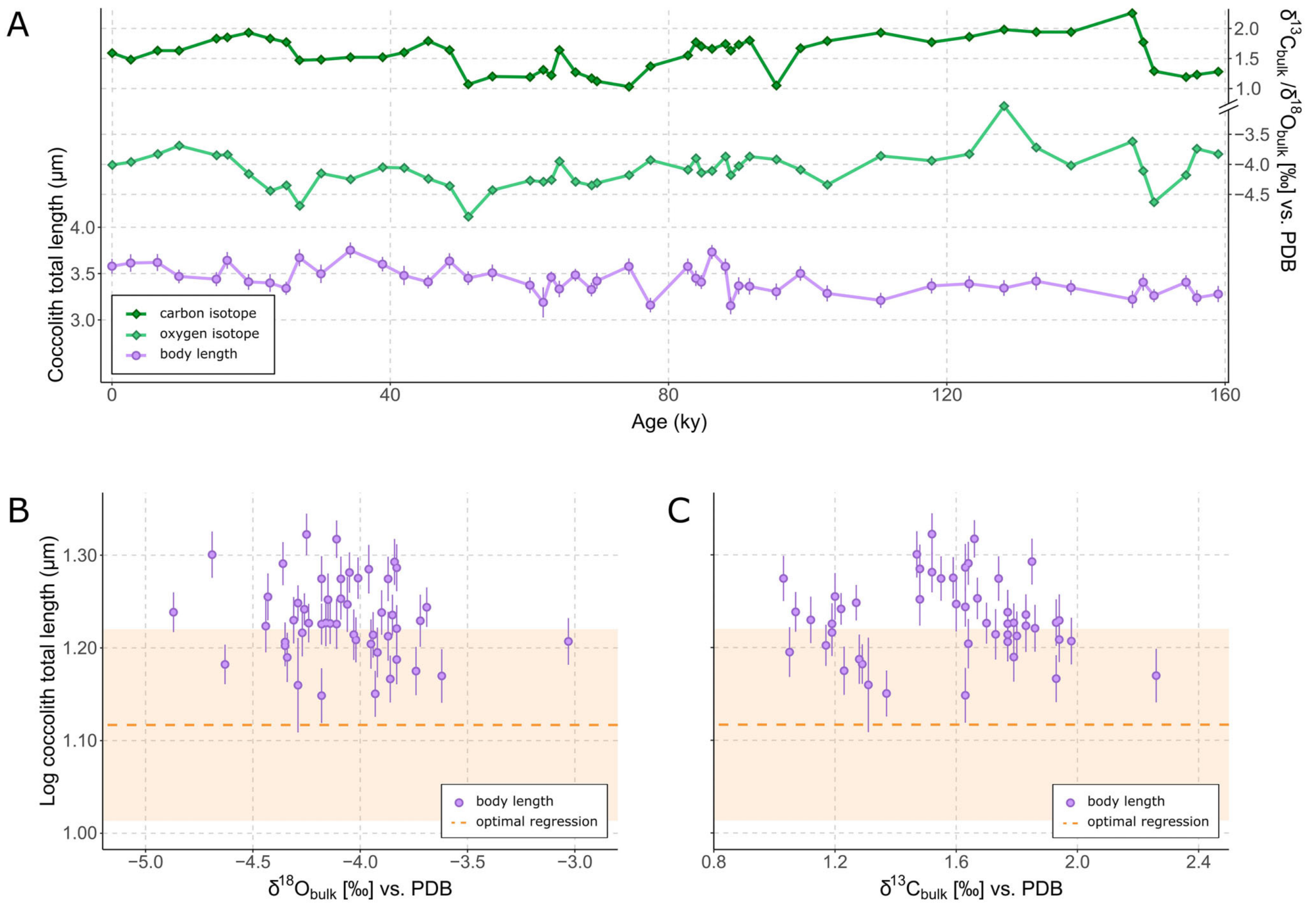

Figure 6. Time series and regressions of the best model (model 3b) for the coccolithophore dataset. (A) Evolution of mean coccolith length (circles) and changes in the environmental proxies (oxygen isotope in dark green diamonds; carbon isotopes in light green diamonds) across time (phenotypic data from Bornemann & Mutterlose, 2006; isotopic data from Bornemann et al., 2005). (B, C) Log coccolith length on oxygen (B) and carbon isotope (C). The optimal trait value and standard error are based on the output of the best-fitting model: $\theta 1$ = 1.12 log(μm2) (—) (an intercept-only regression as the best model does not support a causal relationship between changes in the isotopes and changes in the trait).

fitted did not reach the global likelihood peak in parameter space. While rerunning the analysis with the starting values of the best model improved the relative goodness of the fit, it did not modify the ranking of the models or allow for finding the global likelihood peak for this model. As model 6a describes a causal link between the trait and both environmental variables, estimating the exact value of how these variables affect the optimal coccolith size might run into challenges of parameter non-identifiability due to the collinearity between the two environmental proxies. However, the AICc gap between this model and the best model is large (ΔAICc = 12.68), suggesting this model is unlikely to outcompete the best model even if we detected the global log-likelihood peak.

Foraminifera length

The best model describing body size changes within the foraminifera lineage is the model where all three variables change according to OU processes and where variation in the oxygen isotopes affects the optimal body size (Table 3, model 4a). The optimal regression slope is 0.83, suggesting a strong influence of changes in the oxygen isotopes on the optimal length in the foraminifera lineage: a 10% change in the oxygen isotopes is predicted to lead to 8.3% change in the optimal size of the foraminifera (Figure 7). The half-life is 122 ky while the size of the stochastic changes in the trait is 0.39 log(μm^2), which gives an equilibrium variance of 0.01 log(μm^2). A model recovery analysis confirmed that our modeling framework can identify dependent adaptation with reasonable confidence in the foraminifera system: data simulated from the maximum likelihood estimates of the best-fitting model were correctly classified in 72 of 100 refitting runs (Supplementary material, Table S2).

Two other models have a similar fit relative to the best model. The second-best model (model 6a, ΔAICc = 0.05, Figure S3) returns an optimal regression of 0.74 between the oxygen isotopes and optimal size with a half-life of 73 ky. This model also estimates an optimal regression between changes in the carbon isotopes and foraminifera test size, with a slope of 0.28. The effect of the oxygen isotopic ratio on the trait optimum is therefore substantially stronger compared to the effect of the carbon isotopic ratio. The third best model (model 3a, ΔAICc = 3.28, Figure S4) does not contain any effects of the two isotopes on the optimal size, and size has a substantially longer half-life (562 ky) toward the fixed optimum θ_1 = -1.01 log(μm^2). Both models 3a and 6a recover stochastic changes of a magnitude comparable to

Table 3. Model fit and parameter estimates for the foraminifera dataset.

Model type	Model *(k)*	ΔAICc/logL	Ancestral value ($z_{0.1}, z_{0.2}, z_{0.3}$)	Optimum ($\theta_1, \theta_2, \theta_3$)	A matrix: Diagonal ($\alpha_{11}, \alpha_{22}, \alpha_{33}$)	A matrix: Off-diagonal (α_{12}, α_{13})	Σ matrix: Diagonal ($\sigma_{11}, \sigma_{22}, \sigma_{33}$)	Σ matrix: Off-diagonal ($\sigma_{12}, \sigma_{13}, \sigma_{23}$)
Random evolution	Model 1a *(6)*	9.15/327.15	(-1.49, -1.08, 2.03)	(NA, NA, NA)	(0, 0, 0)	(0, 0)	(0.34, 1.33, 1.28)	(0, 0, 0)
	Model 1b *(9)*	11.44/329.5	(-1.49, -1.08, 2.03)	(NA, NA, NA)	(0, 0, 0)	(0, 0)	(0.35, 1.32, 1.3)	(0, 0.18, 0.18)
Independent adaptation	Model 2a *(8)*	10.34/328.86	(-1.49, -1.08, 2.03)	(-1.01, NA, NA)	(4.18, 0, 0)	(0, 0)	(0.35, 1.33, 1.28)	(0, 0, 0)
	Model 2b *(11)*	12.57/331.38	(-1.49, -1.08, 2.03)	(-0.99, NA, NA)	(4.12, 0, 0)	(0, 0)	(0.35, 1.32, 1.3)	(0, 0.19, 0.18)
	Model 3a *(12)*	3.28/337.28	(-1.49, -1.08, 2.04)	(-1.01, -0.9, 1.69)	(3.95, 9.92, 26.86)	(0, 0)	(0.35, 1.43, 1.56)	(0, 0, 0)
	Model 3b *(15)*	7.36/339.19	(-1.49, -1.08, 2.04)	(-0.99, -0.9, 1.7)	(3.77, 9.4, 27.4)	(0, 0)	(0.35, 1.41, 1.59)	(0.02, 0.18, 0.15)
Dependent adaptation	Model 4a *(13)*	0/340.21	(-1.49, -1.09, 2.04)	(-1.06, -0.85, 1.69)	(18.18, 8.65, 30.57)	(-15.13, 0)	(0.39, 1.27, 1.62)	(0, 0, 0)
	Model 4b *(16)*	6.16/341.16	(-1.5, -1.08, 2.04)	(-1.03, -0.83, 1.71)	(10.94, 9.48, 25.93)	(-7.29, 0)	(0.37, 1.45, 1.55)	(0.02, 0.19, 0.23)
	Model 5a *(13)*	5.86/337.28	(-1.49, -1.08, 2.04)	(-1.03, -0.9, 1.69)	(4.69, 9.82, 27.85)	(0, 0.84)	(0.35, 1.44, 1.59)	(0, 0, 0)
	Model 5b *(16)*	9.44/339.52	(-1.49, -1.08, 2.04)	(-0.99, -0.9, 1.69)	(4.01, 9.99, 26.03)	(0, 2.99)	(0.36, 1.44, 1.56)	(0.03, 0.21, 0.14)
	Model 6a *(14)*	0.05/341.5	(-1.5, -1.08, 2.04)	(-1.11, -0.91, 1.7)	(30.37, 11.91, 35.3)	(-22.55, 8.62)	(0.4, 1.36, 1.71)	(0, 0, 0)
	Model 6b *(17)*	7.92/341.68	(-1.49, -1.08, 2.05)	(-1.09, -0.88, 1.68)	(14.49, 8.91, 24.83)	(-8.17, 2.61)	(0.38, 1.25, 1.46)	(0.02, 0.18, 0.22)

Notes: k is the number of parameters in the model. ΔAICc is the difference in AICc score relative to the best model. z0.1, z0.2, and z0.3 correspond to the ancestral values of the size trait, oxygen, and carbon isotope, while $\theta 1$ (foraminifera length), $\theta 2$ (oxygen proxy), and $\theta 3$ (carbon proxy) are the optimum parameters in the OU process. $\alpha 11$, $\alpha 22$, and $\alpha 33$ represent the diagonal elements of the A matrix, while the off-diagonal elements in the A matrix are $\alpha 12$ and $\alpha 13$. In the Σ matrix, $\sigma 11$, $\sigma 22$, and $\sigma 33$ are the stochastic variances in the size trait, the oxygen isotope, and the carbon isotope, respectively, while $\sigma 12$, $\sigma 13$, and $\sigma 23$ represent the covariances in the random changes in these variables. Standard errors of parameter estimates are available in the Supplementary material (Table S5).

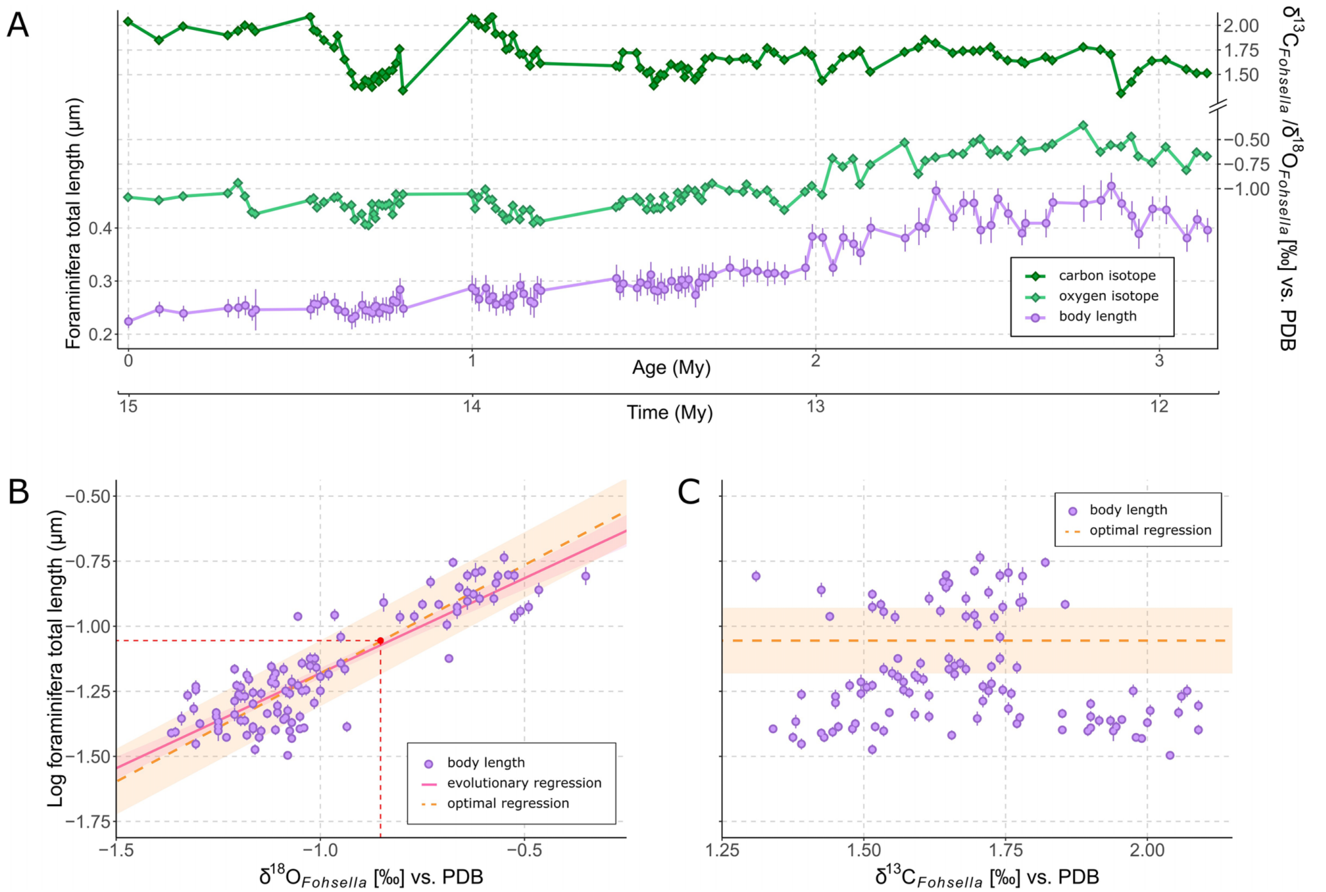

Figure 7. Time series and regressions of the best model (model 4a) for the foraminifera dataset. (A) Evolution of test length mean in the lineage *Globorotalia fohsella* (circle) and changes in the environmental proxies (oxygen isotope in dark green diamonds; carbon isotopes in light green diamonds) across time (data from Hodell & Vayavananda, 1993). (B) Optimal (—) and OLS (—) regression (and standard error) of log foraminifera body length on oxygen isotope based on parameters of the best-fitting model. The slope of the optimal regression is 0.83, while the OLS regression is 0.73. The intercept for the optimal regression is calculated as mean(log body length)–0.83∗mean(oxygen isotope). The variance explained by the optimal regression is R2 = 75%, while it is 76% for the OLS regression. The dot on the optimal regression line represents where the primary optima for the two time series intersect (i.e., the primary optimum for log foraminifera body length when the oxygen isotope has reached its stationary phase). (C) Log foraminifera body length on carbon isotope values. The optimal trait value and standard error are based on the output of the best-fitting model: $\theta1$ = -1.06 log(µm2) (—) (an intercept-only regression as the best model does not support a causal relationship between changes in the carbon isotope and changes in the trait).

the best model and an absence of correlation between those changes.

Discussion

Adaptation within lineages across timescales

We applied multivariate OU models to three time series to test if the adaptation-inertia framework is useful for interpreting trait evolution within lineages across different timescales.

Our results suggest that the decrease in body size in the salmon population in River Eira is a response to shifts of the adaptive optimum caused by decreased water flow in the river. Our results thus align with the conclusion of the original study by Jensen et al. (2022), who found that population genetic models suggested strong selection on body size driven by changes in water flow. Jensen et al. (2022) reported that changes in allele frequencies at two loci (vgll3 and six6) predicted about 80% of the observed change in body mass, suggesting that the change in body size is mainly the result of evolution due to strong selection. The effect of changes in water flow on the optimal trait size is close to unity, and the half-life of 0.91 years indicates very rapid adaptation towards the primary optimum. These results suggest strong selection on body size, supporting hypotheses such as the inaccessibility of the breeding ground for bigger individuals (Jensen et al., 2022) as well as earlier maturation and reduced mortality for smaller individuals (Merilä, 2022) due to reduced water flow in the river. The strong influence of water flow variation on the movements of the primary optimum for salmon body size in River Eira clearly shows that a dynamic adaptive landscape is the preferred model, even across a handful of generations. The ability of the population to track the adaptive peak is aligned with our knowledge of microevolutionary dynamics, where high evolvability (Hansen & Pélabon, 2021) and strong directional selection (Hereford et al., 2004) often result in rapid adaptation (Kinnison & Hendry, 2001). Rapid evolution is also commonly found in systems strongly shaped by human-induced changes (Sanderson et al., 2022). Overall, the results of our analysis of the salmon size data from River Eira reiterate the importance of considering models with a moving optimum also on short timescales, particularly when studying systems undergoing drastic biotic or abiotic changes.

We did not find any convincing evidence supporting that temperature drives size change in the coccolithophore lineage *B. constans*. Colder environments are suggested as a cause for reduced size in *B. constans* in the late Albian (Bornemann & Mutterlose, 2006), but both Lees et al. (2005) and Herrle et al. (2003) described the species as better suited to cool surface water. As concentrations of specimens with reduced size were both recovered during cold and warm periods, Bottini & Faucher (2020) and Erba et al. (2010) hypothesized an indirect relationship between temperature and coccolith size, with ocean acidification acting as a main driver for changes in the trait and isotope proxies. The strong correlation between stochastic variations we recovered in our analysis might suggest that all three variables are influenced by the same set of external factors, and support this latter hypothesis.

We find evidence for adaptation in the foraminifera *Globorotalia (Fohsella)* towards a continuously changing primary optimum due to changes in the oxygen isotope. This corroborates the conclusion in the original study of this data. Hodell & Vayavananda (1993) measured δ^{18}O variations in several lineages of the same samples, but they only recorded an increase in the calcitic shells of the genus *Globorotalia (Fohsella)*, suggesting a decrease in calcification temperature for this specific lineage rather than a fluctuation of continental ice volume. Based on these results, they hypothesized that the increase in δ^{18}O could indicate a gradual vertical migration of the lineage toward a deeper oceanic layer, while the increase in the test length over the same period is interpreted as a buoyancy adjustment. The best-fitting model suggests a direct relationship between changes in temperature and the body size in this foraminifera lineage, aligned with the buoyancy adaptation hypothesized by Hodell & Vayavananda (1993). The second best-fitting model describing these data suggests that δ^{18}O, and also potentially δ^{13}C, are significantly influencing the adaptive optimum of the foraminifera length. In sum, our analyses show that the modeling framework is able to detect evidence of adaptation and a moving topography of the adaptive landscape in data covering millions of years of trait evolution within a lineage. While observation of phenotypic evolution in the fossil record is often found to represent small net change across time (Gingerich, 2001; Uyeda et al., 2011), our results add support to the increasing body of work that has found evidence for continuous tracking of peaks in fossil time series (Holstad et al., 2024; Hunt et al., 2025; Reitan et al., 2012; Voje et al., 2024).

Modeling the dynamics of the adaptive landscape at different timescales

We fitted multivariate OU models to phenotypic time series spanning decadal to million-year intervals to assess evidence of a dynamic adaptive landscape across timescales. However, fitting the same model to data covering different timescales does not necessarily imply that the model, and therefore its parameters, can be interpreted in the same way, regardless of the timescale on which it is applied. For example, if the OU process is interpreted as approximating a population genetic description of how a population is evolving in the neighboring area of a fixed peak in the adaptive landscape (Lande, 1976), model parameters can be interpreted as informative of the curvature of the adaptive peak (i.e., the strength of selection) and the average effective population size in the lineage. Hunt & colleagues (2008 et al. (2008, 2025) fitted OU models to fossil time series covering thousands to tens of thousands of years and made meaningful population genetic interpretations of the OU process. These studies exemplify how combining analyses of high-resolution fossil data with mechanistic models of evolution can increase our understanding of how evolutionary processes operate beyond a handful of generations. Yet, it remains an empirical question across which time intervals, and at which resolution of the samples in a time series, such a population genetic interpretation of the OU process stops making sense.

In this study, we do not interpret the OU process as approximating a population genetic model, but rather apply the adaptation–inertia framework to within-lineage changes across time intervals ranging from decades to millions of years. For the shortest timescale (salmon body size evolution across 76 years), a population genetic interpretation of the OU process does indeed make sense (Jensen et al., 2022), though the observed signal of rapid and directional change in body mass in response to altered river flow is also consistent with adaptive landscape movement under the adaptation-inertia framework. Within this framework, the OU process is interpreted as a description of how the adaptive landscape itself is changing across time. This interpretation is less tightly linked to specific population genetic parameters, but as a consequence, it is not constrained to particular timescales in the way that a population genetic interpretation would be. However, the resolution of the data imposes limits and constraints on how much detail can be uncovered in each case on the dynamics of the adaptive landscape. The strong adaptation signal in the salmon population indicates strong selection to track rapid changes in the primary adaptive peak. Such rapid changes in peaks would, in most cases, be difficult to detect in fossil time series, in which longer timespans typically separate samples. Detected peak movements in fossil time series are thus more likely to reflect cumulative changes in the peaks. The adaptive landscape we model on timescales across thousands and millions of years is therefore more of a global representation of the changes in peak positions (Schluter, 2024). The interpretation of the stochastic part of the model might also be dependent on the timescale of the data. For instance, the stochastic component of the OU model tends to capture variation due to genetic drift on short timescales rather than on long ones, where drift is unlikely to be a major contributor to phenotypic evolution (e.g., Estes & Arnold, 2007). Over longer timescales, the stochastic component may instead be dominated by secondary adaptation and the effects of indirect selection. Nevertheless, applying the adaptation–inertia framework to phenotypic time series can shed light on adaptive processes operating at different timescales, which might help us to better understand phenotypic evolution across the timescale continuum.

A framework to test hypotheses of adaptation within lineages

We used multivariate Ornstein-Uhlenbeck models to assess competing biological hypotheses and test potential drivers of evolution within lineages, as implemented in the R package evoTS (Voje, 2023). A related approach was recently used by Hunt et al. (2025), where the authors used state space models to test for the influence of an environmental variable on trait evolution. Their implementation in the R package paleoTS is univariate, modeling a single focal trait that tracks an exogenous environmental covariate, whereas the multivariate OU models we use treat the trait and the environmental variable jointly. Both approaches can detect drivers of change that go beyond pure correlation; while a correlation is symmetrical (X is correlated with Y the same way as Y is correlated with X), both state space models and multivariate OU processes can find that variable X affects Y in a way that Y does not affect X. A further difference is that in the univariate state space implementation the direction of the effect is imposed by treating the covariate as exogenous, whereas in the multivariate OU framework the direction is itself estimated through the off-diagonal elements of the A matrix. Another related framework is the layered stochastic differential equation approach of Reitan et al. (2012), implemented and generalized in the layeranalyzer R package (Reitan & Liow, 2019), which describes traits tracking optima and uses Granger causality to identify directional effects among processes, much like the multivariate OU models used here. It differs from our approach mainly in that it can explicitly model hidden (unmeasured) layers, reconstructing latent driving processes from the autocorrelation structure of the data. All three approaches share the same underlying linear-Gaussian, causally structured logic, and applying them to within-lineage time series offers complementary routes to linking trait evolution with its hypothesized environmental drivers.

Complexity of the log-likelihood landscape

In evolution models with one or two parameters, like the unbiased random walk (Hunt, 2006), the log-likelihood landscape usually has a single-peak topography, and search algorithms are quickly able to converge on the global peak (see e.g., Kinneberg & Voje, 2025 for a large-scale assessment of this). Finding maximum likelihood estimates of parameters for more complex evolutionary models can be more complicated. For example, if the zone of highest elevation is a ridge, multiple combinations of parameters can have more or less identical log-likelihood (e.g., Ho & Ané, 2014). If the landscape is multi-peaked, search algorithms could converge on a local peak, not recovering the maximum likelihood set of parameters. Finding global likelihood peaks can be challenging for multivariate OU models (e.g., Bartoszek et al., 2023), which we experienced in some of our analyses. Starting the search routine from different positions in the log-likelihood space is therefore recommended in order to better explore the parameter space. We also found it helpful to use the maximum likelihood parameter values of similar models as starting values in order to do a more complete search of the likelihood landscape.

Conclusion

We applied multivariate OU models to three phenotypic time series covering different time intervals and found that models portraying dynamic adaptive landscapes describe phenotypic evolution well both across a handful of generations and over millions of years. Modeling the dynamics of the adaptive landscape using the same modeling framework on data covering timescales that differ in many orders of magnitude may be one way forward in the ongoing process of better connecting evolution across time. We also argue that multivariate OU models are useful tools to assess competing hypotheses of adaptation within lineages.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

Data, scripts, and results are available in a digital repository on GitHub (https://github.com/Marion-Thaureau/Thaureau-Voje-2026) and archived on Dryad (DOI: https://doi.org/10.5061/dryad.95×69p91r)

Author contributions

Both M.T. and K.L.V. conceptualized and designed the project. K.L.V. supervised the project. M.T. analyzed the data, produced the figures, and wrote the original draft with the support of K.L.V. Both M.T. and K.L.V. interpreted the results, contributed to edits and reviews on the manuscript, and gave their approval on the final version.

Funding

The project was supported through an ERC–2020–STG (grant agreement ID: 948465) obtained by K.L.V. under the European Union's Horizon 2020 research and innovation program.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgments

We thank all those who generated the time series data we analyze and made their data publicly available. The phenotypic data are available in the Phenotypic Time Series (PETS) Database (pets.nhm.uio.no/PETS). We also thank two anonymous reviewers for their insightful comments, which improved the manuscript. The computations were performed on resources provided by Sigma2–the National Infrastructure for High-Performance Computing and Data Storage in Norway.

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, *19*(6), 716–723. https://doi.org/10.1109/TAC.1974.1100705

Arnold, S. J., Pfrender, M. E., & Jones, A. G. (2001). The adaptive landscape as a conceptual bridge between micro- and macroevolution. *Genetica*, *112*(1), 9–32. https://doi.org/10.1023/A:1013373907708

Ashton, K. G. (2004). Comparing phylogenetic signal in intraspecific and interspecific body size datasets. *Journal of Evolutionary Biol-*

ogy, *17*(5), 1157–1161. https://doi.org/10.1111/j.1420-9101.2004.00764.x
Auguie, B., & Antonov, A. (2017). *gridExtra: miscellaneous functions for "grid" graphics*. (Version 2.3) [Computer software]. https://cran.r-project.org/web/packages/gridExtra/index.html
Bartoszek, K., Fuentes-González, J., Mitov, V., Pienaar, J., Piwczyński, M., Puchałka, R., Spalik, K., & Voje, K. L. (2023). Model selection performance in phylogenetic comparative methods under multivariate Ornstein–Uhlenbeck models of trait evolution. *Systematic Biology*, *72*(2), 275–293. https://doi.org/10.1093/sysbio/syac079
Bartoszek, K., Pienaar, J., Mostad, P., Andersson, S., & Hansen, T. F. (2012). A phylogenetic comparative method for studying multivariate adaptation. *Journal of Theoretical Biology*, *314*, 204–215. https://doi.org/10.1016/j.jtbi.2012.08.005
Bornemann, A., & Mutterlose, J. (2006). Size analyses of the coccolith species *Biscutum constans* and *Watznaueria barnesiae* from the Late Albian "Niveau Breistroffer" (SE France): Taxonomic and palaeoecological implications. *Geobios*, *39*(5), 599–615. https://doi.org/10.1016/j.geobios.2005.05.005
Bornemann, A., Pross, J., Reichelt, K., Herrle, J. O., Hemleben, C., & Mutterlose, J. (2005). Reconstruction of short-term palaeoceanographic changes during the formation of the Late Albian 'Niveau Breistroffer' black shales (Oceanic Anoxic Event 1d, SE France). *Journal of the Geological Society*, *162*(4), 623–639. https://doi.org/10.1144/0016-764903-171
Bottini, C., & Faucher, G. (2020). *Biscutum constans* coccolith size patterns across the mid Cretaceous in the western Tethys: paleoecological implications. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *555*, 109852. https://doi.org/10.1016/j.palaeo.2020.109852
Brody, D. C., Syroka, J., & Zervos, M. (2002). Dynamical pricing of weather derivatives. *Quantitative Finance*, *2*(3), 189–198. https://doi.org/10.1088/1469-7688/2/3/302
Burnham, K. P., Anderson, D. R., & Huyvaert, K. P. (2011). AIC model selection and multimodel inference in behavioral ecology: Some background, observations, and comparisons. *Behavioral Ecology and Sociobiology*, *65*(1), 23–35. https://doi.org/10.1007/s00265-010-1029-6
Butler, M. A., & King, A. A. (2009). Multivariate comparative analysis using OUCH. *Integrative Comparative Biology*, *49*, E24–E24.
Emiliani, C. (1955). Pleistocene temperatures. *The Journal of Geology*, *63*(6), 538–578. https://doi.org/10.1086/626295
Erba, E., Bottini, C., Weissert, H. J., & Keller, C. E. (2010). Calcareous nannoplankton response to surface-water acidification around Oceanic Anoxic Event 1a. *Science*, *329*(5990), 428–432. https://doi.org/10.1126/science.1188886
Estes, S., & Arnold, S. J. (2007). Resolving the paradox of stasis: models with stabilizing selection explain evolutionary divergence on all timescales. *The American Naturalist*, *169*(2), 227–244. https://doi.org/10.1086/510633
Fairbanks, R. G., Sverdlove, M., Free, R., Wiebe, P. H., & Bé, A. W. H. (1982). Vertical distribution and isotopic fractionation of living planktonic foraminifera from the Panama Basin. *Nature*, *298*(5877), 841–844. https://doi.org/10.1038/298841a0
Fairbanks, R. G., Wiebe, P. H., & Bé, A. W. H. (1980). Vertical distribution and isotopic composition of living planktonic foraminifera in the Western North Atlantic. *Science*, *207*(4426), 61–63. https://doi.org/10.1126/science.207.4426.61
Fleisher, R. L. (1974). Cenozoic planktonic foraminifera and biostratigraphy, Arabian Sea Deep Sea Drilling Project, Leg 23A. In R. B. Whitmarsh (Ed.), *Initial reports of the deep sea drilling project*. (vol. *23*, pp. 1001–1072). U.S. Government Printing Office. https://doi.org/10.2973/dsdp.proc.23.139.1974
Flower, B. P., & Kennett, J. P. (1994). The middle Miocene climatic transition: East Antarctic ice sheet development, deep ocean circulation and global carbon cycling. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *108*(3), 537–555. https://doi.org/10.1016/0031-0182(94)90251-8
Freckleton, R. P., Harvey, P. H., & Pagel, M. (2002). Phylogenetic analysis and comparative data: a test and review of evidence. *The American Naturalist*, *160*(6), 712–726. https://doi.org/10.1086/343873
Gingerich, P. D. (2001). Rates of evolution on the time scale of the evolutionary process. In A. P. Hendry, & M. T. Kinnison (Eds.), *Microevolution rate, pattern, process*. (pp. 127–144). Springer. https://doi.org/10.1007/978-94-010-0585-2_9
Gingerich, P. D. (2009). Rates of evolution. *Annual Review of Ecology, Evolution, and Systematics*, *40*(1), 657–675. https://doi.org/10.1146/annurev.ecolsys.39.110707.173457
Grabowski, M., Pienaar, J., Voje, K. L., Andersson, S., Fuentes-González, J., Kopperud, B. T., Moen, D. S., Tsuboi, M., Uyeda, J., & Hansen, T. F. (2023). A cautionary note on "A cautionary note on the use of Ornstein Uhlenbeck models in macroevolutionary studies. *Systematic Biology*, *72*(4), 955–963. https://doi.org/10.1093/sysbio/syad012
Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, *37*, 424–438. https://doi.org/10.2307/1912791
Grant, P. R., & Grant, B. R. (2002). Unpredictable evolution in a 30-year study of Darwin's finches. *Science*, *296*(5568), 707–711. https://doi.org/10.1126/science.1070315
Grant, P. R., & Grant, B. R. (2006). Evolution of character displacement in Darwin's finches. *Science*, *313*(5784), 224–226. https://doi.org/10.1126/science.1128374
Hansen, T. F. (1997). Stabilizing selection and the comparative analysis of adaptation. *Evolution*, *51*(5), 1341–1351. https://doi.org/10.1111/j.1558-5646.1997.tb01457.x
Hansen, T. F. (2012). Adaptive landscapes and macroevolutionary dynamics. In E. Svensson, & R. Calsbeek (Eds.), *The adaptive landscape in evolutionary biology*. (pp. 205–226). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199595372.003.0013
Hansen, T. F., & Houle, D. (2008). Measuring and comparing evolvability and constraint in multivariate characters. *Journal of Evolutionary Biology*, *21*(5), 1201–1219. https://doi.org/10.1111/j.1420-9101.2008.01573.x
Hansen, T. F., & Pélabon, C. (2021). Evolvability: a quantitative-genetics perspective. *Annual Review of Ecology, Evolution, and Systematics*, *52*(1), 153–175. https://doi.org/10.1146/annurev-ecolsys-011121-021241
Hansen, T. F., Pienaar, J., & Orzack, S. H. (2008). A comparative method for studying adaptation to a randomly evolving environment. *Evolution*, *62*(8), 1965–1977. https://doi.org/10.1111/j.1558-5646.2008.00412.x
Hereford, J., Hansen, T. F., & Houle, D. (2004). Comparing strengths of directional selection: how strong is strong? *Evolution*, *58*(10), 2133–2143. https://doi.org/10.1111/j.0014-3820.2004.tb01592.x
Herrle, J.O., Pross, J., Friedrich, O., Kößler, P., & Hemleben, C. (2003). Forcing mechanisms for mid-Cretaceous black shale formation: Evidence from the Upper Aptian and Lower Albian of the Vocontian Basin (SE France). *Palaeogeography, Palaeoclimatology, Palaeoecology*, *190*, 399–426. https://doi.org/10.1016/S0031-0182(02)00616-8
Ho, L. S. T., & Ané, C. (2014). Intrinsic inference difficulties for trait evolution with Ornstein-Uhlenbeck models. *Methods in Ecology and Evolution*, *5*(11), 1133–1146. https://doi.org/10.1111/2041-210X.12285
Hodell, D. A., & Vayavananda, A. (1993). Middle Miocene paleoceanography of the western equatorial Pacific (DSDP site 289) and the evolution of *Globorotalia (Fohsella)*. *Marine Micropaleontology*, *22*(4), 279–310. https://doi.org/10.1016/0377-8398(93)90019-T
Holstad, A., Voje, K. L., Opedal, Ø. H., Bolstad, G. H., Bourg, S., Hansen, T. F., & Pélabon, C. (2024). Evolvability predicts macroevolution under fluctuating selection. *Science*, *384*(6696), 688–693. https://doi.org/10.1126/science.adi8722

Houle, D., Pélabon, C., Wagner, G. P., & Hansen, T. F. (2011). Measurement and meaning in biology. *The Quarterly Review of Biology*, *86*(1), 3–34. https://doi.org/10.1086/658408

Hunt, G. (2006). Fitting and comparing models of phyletic evolution: Random walks and beyond. *Paleobiology*, *32*(4), 578–601. https://doi.org/10.1666/05070.1

Hunt, G. (2007). The relative importance of directional change, random walks, and stasis in the evolution of fossil lineages. *Proceedings of the National Academy of Sciences*, *104*(47), 18404–18408. https://doi.org/10.1073/pnas.0704088104

Hunt, G., Bell, M. A., & Travis, M. P. (2008). Evolution toward a new adaptive optimum: phenotypic evolution in a fossil stickleback lineage. *Evolution*, *62*(3), 700–710. https://doi.org/10.1111/j.1558-5646.2007.00310.x

Hunt, G., Hopkins, M. J., & Lidgard, S. (2015). Simple versus complex models of trait evolution and stasis as a response to environmental change. *Proceedings of the National Academy of Sciences*, *112*(16), 4885–4890. https://doi.org/10.1073/pnas.1403662111

Hunt, G., Martínez-Rivera, W., Hopkins, M. J., Fricks, J., & Sterner, B. (2025). Using state space models to understand trait evolution in fossil lineages. *Paleobiology*, 1–15. https://doi.org/10.1017/pab.2025.10075

Hunt, G., & Roy, K. (2006). Climate change, body size evolution, and Cope's Rule in deep-sea ostracodes. *Proceedings of the National Academy of Sciences*, *103*(5), 1347–1352. https://doi.org/10.1073/pnas.0510550103

Hunt, G., Wicaksono, S. A., Brown, J. E., & Macleod, K. G. (2010). Climate-driven body-size trends in the ostracod fauna of the deep Indian Ocean. *Palaeontology*, *53*(6), 1255–1268. https://doi.org/10.1111/j.1475-4983.2010.01007.x

Jensen, A. J., Hagen, I. J., Czorlich, Y., Bolstad, G. H., Bremset, G., Finstad, B., Hindar, K., Skaala, Ø., & Karlsson, S. (2022). Large-effect loci mediate rapid adaptation of salmon body size after river regulation. *Proceedings of the National Academy of Sciences*, *119*(44), e2207634119. https://doi.org/10.1073/pnas.2207634119

Kinneberg, V. B., & Voje, K. L. (2025). Rate-time scaling in phenotypic evolution: limitations of current models in capturing temporal dynamics. *Evolution*, qpaf208. https://doi.org/10.1093/evolut/qpaf208

Kinnison, M. T., & Hendry, A. P. (2001). The pace of modern life II: From rates of contemporary microevolution to pattern and process. *Genetica*, *112*(1), 145–164. https://doi.org/10.1023/A:1013375419520 .

Lande, R. (1976). Natural selection and random genetic drift in phenotypic evolution. *Evolution*, *30*(2), 314–334. https://doi.org/10.1111/j.1558-5646.1976.tb00911.x

Lande, R. (1979). Quantitative genetic analysis of multivariate evolution, applied to brain: body size allometry. *Evolution*, *33*(1), 402. https://doi.org/10.2307/2407630

Lande, R., & Arnold, S. J. (1983). The measurement of selection on correlated characters. *Evolution*, *37*(6), 1210. https://doi.org/10.2307/2408842

Lees, J. A., Brown, P. R., & Mattioli, E. (2005). Problems with proxies? Cautionary tales of calcareous nannofossil paleoenvironmental indicators. *Micropaleontology*, *51*(4), 333–343. https://doi.org/10.2113/gsmicropal.51.4.333

Losos, J. B., Warheitt, K. I., & Schoener, T. W. (1997). Adaptive differentiation following experimental island colonization in Anolis lizards. *Nature*, *387*(6628), 70–73. https://doi.org/10.1038/387070a0

Lynch, M., & Walsh, J. B. (1998). *Genetics and analysis of quantitative traits*. Sinauer Associates.

Merilä, J. (2022). Human-induced evolution of salmon by means of unnatural selection. *Proceedings of the National Academy of Sciences*, *119*(44), e2216526119. https://doi.org/10.1073/pnas.2216526119

Meyer, A. (1987). Phenotypic plasticity and heterochrony in cichlasoma managuense (pisces, cichlidae) and their implications for speciation in cichlid fishes. *Evolution*, *41*(6), 1357–1369. https://doi.org/10.1111/j.1558-5646.1987.tb02473.x

Mitov, V., Bartoszek, K., & Stadler, T. (2019). Automatic generation of evolutionary hypotheses using mixed Gaussian phylogenetic models. *Proceedings of the National Academy of Sciences*, *116*(34), 16921–16926. https://doi.org/10.1073/pnas.1813823116

Pienaar, J., Bartoszek, K., Brahmantio, B., Fierst, J. L., Fuentes-González, J. A., Hansen, T. F., Kiang, W. H. C., Kopperud, B. T., & Voje, K. L. (2026). Phylogenetic comparative methods for studying adaptation: The adaptation-inertia framework. *Journal of Evolutionary Biology*, *39*(1), 1–17. https://doi.org/10.1093/jeb/voaf113

Potts, R. (1998). Environmental hypotheses of hominin evolution. *American Journal of Physical Anthropology*, *107*(S27), 93–136. https://doi.org/10.1002/(SICI)1096-8644(1998)107:27+⟨93::AID-AJPA5⟩3.0.CO;2-X

Ravelo, A. C., & Hillaire-Marcel, C. (2007). The use of oxygen and carbon isotopes of foraminifera in paleoceanography. In C. Hillaire–Marcel, & A. de Vernal (Eds.), *Developments in marine geology*. (Vol. *1*, pp. 735–764). Elsevier. https://doi.org/10.1016/S1572-5480(07)01023-8

Reitan, T., & Liow, L. H. (2019). layeranalyzer: Inferring correlative and causal connections from time series data in r. *Methods in Ecology and Evolution*, *10*(12), 2183–2188. https://doi.org/10.1111/2041-210X.13299

Reitan, T., & Petersen-Øverleir, A. (2011). Dynamic rating curve assessment in unstable rivers using Ornstein-Uhlenbeck processes. *Water Resources Research*, *47*(2), 2010WR009504. https://doi.org/10.1029/2010WR009504

Reitan, T., Schweder, T., & Henderiks, J. (2012). Phenotypic evolution studied by layered stochastic differential equations. *The Annals of Applied Statistics*, *6*(4), 1531–1551. https://doi.org/10.1214/12-AOAS559

Rolland, J., Henao-Diaz, L. F., Doebeli, M., Germain, R., Harmon, L. J., Knowles, L. L., Liow, L. H., Mank, J. E., Machac, A., Otto, S. P., Pennell, M., Salamin, N., Silvestro, D., Sugawara, M., Uyeda, J., Wagner, C. E., & Schluter, D. (2023). Conceptual and empirical bridges between micro- and macroevolution. *Nature Ecology & Evolution*, *7*(8), 1181–1193. https://doi.org/10.1038/s41559-023-02116-7

Sanderson, S., Beausoleil, M. -O., O'Dea, R. E., Wood, Z. T., Correa, C., Frankel, V., Gorné, L. D., Haines, G. E., Kinnison, M. T., Oke, K. B., Pelletier, F., Pérez-Jvostov, F., Reyes-Corral, W. D., Ritchot, Y., Sorbara, F., Gotanda, K. M., & Hendry, A. P. (2022). The pace of modern life, revisited. *Molecular Ecology*, *31*(4), 1028–1043. https://doi.org/10.1111/mec.16299

Schluter, D. (2024). Variable success in linking micro- and macroevolution. *Evolutionary Journal of the Linnean Society*, *3*(1), kzae016. https://doi.org/10.1093/evolinnean/kzae016

Simon, M. N., Courtois, E. A., Herrel, A., & Moen, D. S. (2025). Macroevolutionary divergence along allometric lines of least resistance in frog hindlimb traits and its effect on locomotor evolution. *The American Naturalist*, *205*(6), 637–655. https://doi.org/10.1086/735569

Simpson, G. G. (1944). *Tempo and mode in evolution*. Colombia University Press.

Simpson, G. G. (1953). *The major features of evolution*. Columbia University Press. https://doi.org/10.7312/simp93764

Toljagić, O., Voje, K. L., Matschiner, M., Liow, L. H., & Hansen, T. F. (2018). Millions of years behind: slow adaptation of ruminants to grasslands. *Systematic Biology*, *67*(1), 145–157. https://doi.org/10.1093/sysbio/syx059

Tsuboi, M., Kopperud, B. T., Matschiner, M., Grabowski, M., Syrowatka, C., Pélabon, C., & Hansen, T. F. (2024). Antler allometry, the irish elk and Gould revisited. *Evolutionary Biology*, *51*(1), 149–165. https://doi.org/10.1007/s11692-023-09624-1

Uyeda, J. C., Hansen, T. F., Arnold, S. J., & Pienaar, J. (2011). The million-year wait for macroevolutionary bursts. *Proceedings of the National Academy of Sciences*, *108*(38), 15908–15913. https://doi.org/10.1073/pnas.1014503108

Voje, K. L. (2016). Tempo does not correlate with mode in the fossil record: tempo and mode in evolution. *Evolution*, *70*(12), 2678–2689. https://doi.org/10.1111/evo.13090

Voje, K. L. (2023). Fitting and evaluating univariate and multivariate models of within-lineage evolution. *Paleobiology*, *49*(4), 747–764. https://doi.org/10.1017/pab.2023.10

Voje, K. L., & Hansen, T. F. (2013). Evolution of static allometries: adaptive change in allometric slopes of eye span in stalk-eyed flies. *Evolution*, *67*(2), 453–467. https://doi.org/10.1111/j.1558-5646.2012.01777.x

Voje, K. L., Saito-Kato, M., & Spanbauer, T. L. (2024). Evolution in fossil time series reconciles observations in micro- and macroevolution. *Journal of Evolutionary Biology*, *37*(12), 1551–1562. https://doi.org/10.1093/jeb/voae087

Voje, K. L., Saulsbury, J. G., Starrfelt, J., Latorre, D. V., Rojas, A., Kinneberg, V. B., Liow, L. H., Wilson, C. J., Saupe, E. E., & Grabowski, M. (2023). Measurement theory and paleobiology. *Trends in Ecology & Evolution*, *38*(12), 1165–1176. https://doi.org/10.1016/j.tree.2023.08.005

Voje, K. L., Starrfelt, J., & Liow, L. H. (2018). Model adequacy and microevolutionary explanations for stasis in the fossil record. *The American Naturalist*, *191*(4), 509–523. https://doi.org/10.1086/696265

Walsh, B., & Lynch, M. (2018). *Evolution and selection of quantitative traits*. Oxford University Press. https://doi.org/10.1093/oso/9780198830870.001.0001

Wickham, H. (2016). *Ggplot2: elegant graphics for data analysis*. Springer-Verlag . https://doi.org/10.1007/978-3-319-24277-4

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *dplyr: a grammar of data manipulation*. (Version 1.1.4) [Computer software] (Accessed 20 January 2024). https://cran.r-project.org/web/packages/dplyr/index.html

Wickham, H., Pedersen, T. L., & Seidel, D. (2025). *scales: scale functions for visualization*. (Version 1.4.0) [Computer software] (Accessed 15 August 2025). https://cran.r-project.org/web/packages/scales/index.html

Woodruff, F., & Savin, S. M. (1989). Miocene deepwater oceanography. *Paleoceanography*, *4*(1), 87–140. https://doi.org/10.1029/PA004i001p00087

Xu, S., Chen, M., Feng, T., Zhan, L., Zhou, L., & Yu, G. (2021). Use ggbreak to effectively utilize plotting space to deal with large datasets and outliers. *Frontiers in Genetics*, *12*, 774846. https://doi.org/10.3389/fgene.2021.774846

Received January 14, 2026; revisions received June 19, 2026; accepted June 29, 2026
Associate Editor: Laura Alencar; Handling Editor: Hélène Morlon